

Fokrul Islam Bhuiyan

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SUMMARY

I have a strong background in software engineering, with a focus on automation, AI, and Machine Learning. Key highlights include publishing research in Scientific Reports and Computers in Biology and Medicine, and developing AI models using YOLO for real-time object detection. My expertise extends to data engineering initiatives, building automated pipelines, and performing large-scale data analysis to drive client insights. Additionally, my experience as an SQA engineer has honed my skills in designing test plans, managing defect lifecycles, and optimizing testing processes through innovative approaches. This unique combination of expertise in AI, data engineering, and software testing positions me to make meaningful contributions to data science and ML engineering.

TECHNICAL SKILLS

- **Deep Learning & GenAI:** LLMs, RAG Pipelines, LangChain, Transformers, Transfer Learning, YOLO, Object Detection, Instance Segmentation, Image Processing, Federated learning (FL).
- **Frameworks & Languages:** Django, Flask, JavaScript, FastAPI, Streamlit.
- **Data Engineering:** Web Scraping, Data Pipelines, Feature Engineering, SQL.
- **QA & Automation:** mabl, Selenium, Manual Testing, Test Planning, Defect Lifecycle.
- **Deployment:** Docker, AWS.
- **Analysis:** Python, Tableau, Matplotlib, Statistical Modeling.

PUBLICATIONS

- **Scientific Reports (2025):** "Enhancing the YOLOv8 model for realtime object detection to ensure online platform safety."

- **Computers in Biology and Medicine (Accepted/In Press - 2026):** "WhyMedQA: Enhanced biomedical why question answering using transfer learning approach."

EXPERIENCE

Data Scientist (Freelance/Contract) | Fiverr | April 2025 – Present

- **AI Model Development & Deployment:** Built and deployed production-ready solutions, including **RAG-based LLM applications** for NLP tasks and **YOLO** models for real-time computer vision segmentation.
- **Data Engineering Pipelines:** Engineered automated web scraping pipelines to harvest large-scale datasets and performed advanced data analysis to drive client decision-making.
- **End-to-End Implementation:** Managed the full project lifecycle from raw data preprocessing to model training and cloud deployment via **FastAPI** and **Docker**.

SQA engineer | Tekarsh Bangladesh Limited | Aug 2021 – Present

- **Test Planning & Defect Management:** Designed early-stage test plans based on requirement analysis to detect **critical bugs** pre-release, ensuring high stability for production builds.
- **Pilot Initiatives & Process R&D:** Executed **pilot projects** to assess the feasibility of new QA strategies—ranging from operational workflows to automation tools (e.g., **mabl**, **Claude**)—optimizing these processes for efficiency before team-wide implementation.
- **Team Leadership & Mentoring:** Mentored junior engineers on automation best practices and root cause analysis, conducting training sessions to foster a collaborative culture of quality improvement.

Teacher assistant (internship) | American International University-Bangladesh | Sep 2020 – Jan 2021

- **Instructional Support:** Facilitated dynamic learning experiences by assisting in lesson delivery and monitoring student engagement, utilizing diverse instructional techniques to support learning goals.
- **Performance Assessment:** Managed grading verification and academic evaluation processes, ensuring 100% accuracy in student performance records and providing constructive feedback.

EDUCATION

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| 2017 – 2021 | B.Sc. in Software Engineering
<i>American International University-Bangladesh</i> |
| 2014 – 2016 | Higher Secondary School Certificate in Science
<i>Ibn Taimiya School and College, Cumilla, Bangladesh</i> |
| 2008 – 2014 | Secondary School Certificate in Science
<i>Ibn Taimiya School and College, Cumilla, Bangladesh</i> |

CERTIFICATIONS

- **Deep Learning Specialization** – Coursera (Andrew Ng)
 - Convolutional Neural Networks
 - Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
 - Neural Networks and Deep Learning
- **Data Analysis with Python** – Coursera (IBM)
 - Data Analysis with Python
- **Problem Solving (Basic)** – HackerRank
 - Problem Solving (Basic)

LANGUAGES

- English: Intermediate
- Bangla: Native